

On Some Tempting but Insufficient Routes to Fixed-Radical Kummer Non-Concentration

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Abstract

We record several plausible but insufficient routes to a fixed-radical Kummer non-concentration theorem arising in a conditional approach to Erdős–Graham Problem #203. The purpose is defensive and clarifying: to separate correct structural lemmas from missing analytic distribution estimates. The note identifies the high- ℓ prime Hecke non-concentration seam as the precise remaining analytic problem.

1 The target theorem

For $\ell \leq (\log Q)^B$ and $(r, s) \neq (0, 0) \in \mathbb{F}_\ell^2$, define

$$\theta = 2^r 3^s.$$

The desired non-concentration bound is

$$\#\{q \leq Q : q \equiv 1 \pmod{\ell}, \theta \in (\mathbb{F}_q^*)^\ell\} \leq (1 - \ell^{-C})\pi(Q; \ell, 1).$$

This is PHNC.

2 Correct structural facts

2.1 Kummer independence

For $(r, s) \neq 0$, the element $2^r 3^s$ is not an ℓ -th power in $\mathbb{Q}(\zeta_\ell)$. This follows from valuations at primes above 2 and 3. Therefore the Kummer layer

$$\mathbb{Q}(\zeta_\ell, (2^r 3^s)^{1/\ell})/\mathbb{Q}(\zeta_\ell)$$

has degree ℓ .

2.2 Order and non-real character

The corresponding Kummer character has order exactly ℓ . If $\ell \geq 3$, it is non-real.

2.3 Power-stability of small deficit

Let $f : X \rightarrow S^1$ and

$$\delta(f) = 1 - |\mathbb{E}f|.$$

Then

$$\delta(f^m) \leq m^2 \delta(f).$$

Consequently, if an order- ℓ character has deficit $\leq \ell^{-C}$, $C \geq 3$, then its entire nontrivial cyclic line has deficit $\leq 1/2$ for large ℓ .

3 Tempting route 1: BFI as a bad-set bound

A tempting claim is that Bombieri–Friedlander–Iwaniec type results imply a bound of the shape

$$|B_{1/2}| \ll \frac{\log Q}{\ell}$$

for the set of Kummer characters with deficit at most $1/2$.

Warning 1. *This is not a valid plug-in citation. The BFI theorem concerns primes in arithmetic progressions with well-factorable weights. At minimum, it is not a plug-in source for this fixed-radical Hecke/Kummer bad-character bound uniformly in ℓ ; any use would require a precise theorem statement matching this family.*

3.1 What the bound would imply if true

Proposition 1. *Suppose one had*

$$|B_{1/2}| \leq C_0 \frac{\log Q}{\ell}.$$

Then in the regime

$$\ell^2 > C_0 \log Q \cdot \frac{\ell}{\ell - 1},$$

PHNC would follow.

Proof. If PHNC failed, there would be a character χ with $\delta(\chi) \leq \ell^{-C}$. The power-stability lemma implies that the $\ell - 1$ nontrivial powers $\chi, \dots, \chi^{\ell-1}$ all lie in $B_{1/2}$. Hence $|B_{1/2}| \geq \ell - 1$. The assumed bound gives $|B_{1/2}| < \ell - 1$, contradiction. $\square$

The structural part is correct. The missing part is the analytic bad-set bound.

4 Tempting route 2: Plünnecke–Ruzsa

Another tempting route is to use additive combinatorics on the set of bad characters.

Warning 2. *Plünnecke–Ruzsa does not by itself provide prime distribution. In this setting, the one-character cyclic-line implication is already obtained by the elementary L^2 power-stability lemma. The hard part is still the analytic bound on how many medium-deficit characters exist.*

5 Tempting route 3: fixed-field ineffective Chebotarev

For each fixed ℓ, r, s , Chebotarev gives eventual equidistribution:

$$\#\{q \leq Q : q \equiv 1 \pmod{\ell}, \theta \in (\mathbb{F}_q^*)^\ell\} \sim \frac{1}{\ell} \pi(Q; \ell, 1).$$

Warning 3. *This does not by itself imply PHNC uniformly for the growing family*

$$\ell \leq (\log Q)^B.$$

An ineffective threshold depending on the individual field does not automatically yield a single $Q_0(B, C)$ valid for all $\ell \leq (\log Q)^B$.

6 Tempting route 4: multiplicative-order lower bounds

One might try to use lower bounds on $\text{ord}_q(2^r 3^s)$.

Warning 4. *The condition that $2^r 3^s$ is an ℓ -th power modulo q is equivalent to*

$$\text{ord}_q(2^r 3^s) \mid \frac{q-1}{\ell}.$$

For polylogarithmic ℓ , the upper bound $(q-1)/\ell$ is still very large. General lower bounds for multiplicative order do not rule out this divisibility for almost all $q \equiv 1 \pmod{\ell}$.

7 Tempting route 5: generic Chebotarev–Brun–Titchmarsh

Chebotarev–Brun–Titchmarsh type estimates are often useful for upper bounds on splitting primes.

Warning 5. *Known forms carry field-invariant dependencies. In the present growing cyclotomic/Kummer family, those dependencies again restrict the usable range. They do not automatically prove the full high- ℓ seam*

$$(\log Q)^{1-\varepsilon} < \ell \leq (\log Q)^B.$$

8 The actual open seam

The remaining analytic problem is:

Problem 1 (High-seam fixed-radical Kummer non-concentration). *For every fixed $B > 0$, prove that there exists $C = C(B) > 0$ such that, uniformly for all primes*

$$\ell \leq (\log Q)^B$$

and all nonzero $(r, s) \in \mathbb{F}_\ell^2$,

$$\#\{q \leq Q : q \equiv 1 \pmod{\ell}, 2^r 3^s \in (\mathbb{F}_q^*)^\ell\} \leq (1 - \ell^{-C}) \pi(Q; \ell, 1).$$

9 Tone and purpose

This note should be read as an audit ledger, not as a criticism of the cited theorems. The issue in each case is not that the classical theorem is weak or wrong; the issue is that the theorem does not match the exact variable, family, or uniformity requirement of high-seam PHNC without additional work.

10 What survived the failed routes

The failure of the PHNC close does not invalidate the following components:

1. Kummer independence of $2^r 3^s$;
2. nontriviality and exact order ℓ of the Kummer characters;
3. the power-stability lemma;
4. the bad cyclic-line consequence;
5. the projective slope interpretation;
6. the exact identification of the high- ℓ seam.

Thus the audit separates a correct structural component from a missing analytic distribution component. That separation is the main value of the note.

11 No-go theorem template

For every proposed proof route, record:

1. the claimed input;
2. the exact theorem actually available;
3. the mismatch in variables, range, or family;
4. what the input would imply if true;
5. current status.

This note is an audit ledger, not a claim that no future theorem can close the seam.

12 Conclusion

The failed routes are useful because they isolate the exact gap. The structural algebra is real; the high- ℓ analytic distribution theorem remains open.

References

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